

RESEARCH PROJECT DEFINITION

Physics-Informed Reduced-Order Modeling for Inverse Stenosis Analysis

*Technical framework: first-principles baseline with a geometry-aware
POD-Galerkin PINN discrepancy correction*

Independent Research Project

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Project definition and planning document. This is not a clinical protocol, diagnostic system, or claim of validated patient-specific performance.

Document Control

Revision History

Version	Date	Summary
0.1	10 July 2026	Initial project definition centered on a full-field geometry-aware POD-Galerkin PINN with a later residual closure.
0.2	11 July 2026	Reframed the method around a first-principles baseline plus a reduced discrepancy correction; added the virtual-cohort strategy; rewrote the document for clearer cross-disciplinary reading.

Document Purpose and Use

This document defines the current research idea, the questions that will be tested, the planned model structure, the expected data needs, and the limits of any claims. It is written for engineers, researchers, and faculty members who need a clear overview of the project without reading the full engineering notebook.

The document describes a proposed method, not a completed result. It explains why the approach may be useful and also identifies the tests that could show that it is not. The engineering notebook will hold dated thoughts, source evaluations, experiments, failed approaches, code decisions, and detailed changes. A later technical report or paper, if supported by results, would describe the final implementation and evidence.

Document Version Policy

The repository should keep one clearly named current version and archive only versions that mark a meaningful change in the research direction. Git history already preserves minor edits. Version 0.1 should be retained because version 0.2 records a real architectural pivot; ordinary wording, citation, and formatting corrections should update the current file rather than create a new numbered document.

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Executive Summary

Arterial stenosis changes how blood accelerates, loses pressure, and separates from the vessel wall. Computational fluid dynamics (CFD) can calculate these effects in detail, but it becomes expensive when many geometries, flow conditions, or inverse estimates must be evaluated. Simpler analytical and one-dimensional models are much faster and often describe the main pressure-flow behavior well, but they cannot resolve every local feature near a severe or irregular narrowing [2,3,17].

This project proposes a hybrid model that divides the work between those two levels. A low-cost first-principles model will provide the main hemodynamic prediction. A reduced learned model will then predict the part that the baseline misses. The correction will be built from differences between the baseline and higher-fidelity velocity and pressure fields. Proper orthogonal decomposition (POD) will compress those differences into a small set of spatial patterns, and a POD-Galerkin physics-informed neural network (PINN) will estimate the corresponding coefficients and selected stenosis parameters [1,4-6].

$$\text{predicted field} = \text{first-principles baseline} + \text{reduced discrepancy correction} \quad (1)$$

The inverse problem is the main application. Given a limited set of measurements, the model will attempt to estimate stenosis severity and, in later stages, lesion length, location, or other geometric and physiological parameters. The primary reconstructed fields will be velocity and pressure. Wall shear stress, pressure drop, and recirculation measures will normally be calculated from those fields rather than treated as unrelated neural-network outputs.

A second part of the project is a physiologically constrained virtual cohort. Published population statistics, authorized de-identified data where available, and first-principles simulation will be used to generate plausible combinations of vessel geometry, blood pressure, flow, rheology, stenosis properties, and missing measurements. The variables will be generated jointly and filtered by physical rules; hemodynamic outputs will come from the model hierarchy rather than being assigned independently. This follows the general idea of existing virtual arterial populations while extending it toward a reduced discrepancy model and inverse field reconstruction [7,8].

The project is an investigation, not a claim that the proposed architecture is automatically better. The discrepancy may not be easier to compress than the full flow field, the chosen baseline may be too crude, and sparse measurements may not uniquely identify a lesion. Full-field POD models, data-only neural networks, and the uncorrected first-principles model will remain comparison cases. The intended outcome is a verified open-source research implementation and a clear account of where the method works, where it does not, and what evidence would be required before any patient-specific use could be considered.

1. Project Definition and Research Questions

1.1 Current project status

This is an independent research and software project developed from public literature, public or authorized data, and personally available computing resources. The research direction is defined, but the exact baseline model, data source, geometry family, and learning architecture remain open until the first feasibility tests are completed. The project is separate from the author's aerospace thesis work.

1.2 Central research idea

Working hypothesis

A low-cost first-principles hemodynamic model, corrected by a geometry-aware POD-Galerkin PINN trained on the remaining velocity and pressure discrepancy, may support faster forward and inverse stenosis analysis than repeated high-fidelity CFD while retaining useful accuracy inside a clearly defined parameter range.

The idea contains three questions that must be kept separate. First, can the baseline describe enough of the flow that the remaining error is compact? Second, can the reduced correction recover that error with less data or computation than a direct full-field surrogate? Third, can the combined model infer unknown stenosis parameters from the measurements that would realistically be available? A positive result in one part does not guarantee success in the others.

1.3 Primary research questions

- Can the difference between an inexpensive first-principles model and a higher-fidelity stenotic-flow solution be represented with fewer POD modes than the full velocity and pressure fields?
- Can a trained baseline-plus-correction model reconstruct velocity and pressure and estimate stenosis severity at a useful online speedup relative to the selected high-fidelity reference?
- Which pressure, flow, velocity, or imaging-derived measurements contain enough information to identify stenosis severity, and when is the inverse problem ambiguous?
- Does reduced physical structure lower the amount of high-fidelity training data required compared with data-only reduced models?
- How do geometry variation, missing measurements, measurement noise, and uncertainty in flow conditions affect the result?

1.4 Intended contribution

The contribution sought is an engineering implementation and evaluation of several ideas that have usually been studied separately. POD-Galerkin PINNs have been demonstrated for inverse Navier-Stokes problems [1]. Geometry-informed snapshots have been used for blood-flow ROMs on related but different vessel shapes [6]. Virtual arterial populations have been used to study disease detection from hemodynamic measurements [7,8]. This project asks whether those directions can be combined into one physics-informed correction model for inverse stenosis analysis.

- A clear baseline-plus-discrepancy formulation for stenotic velocity and pressure fields.
- A geometry-compatible POD representation of the discrepancy across a controlled stenosis family.
- An inverse reduced model for stenosis severity and later geometric parameters.
- A virtual-cohort method that keeps geometry, physiology, disease parameters, and simulated measurements physically consistent.
- Direct comparisons with simpler full-field, data-only, and uncorrected models.
- A reproducible software implementation and, after validation, a limited demonstration interface.

1.5 Meaning of a useful negative result

The project remains worthwhile if it shows that the proposed split is not useful. Examples include a discrepancy that needs nearly as many modes as the full field, a baseline that creates artificial correction structure, or sparse measurements that cannot distinguish between several lesions. Those findings would define where this model should not be used and may support a smaller comparison study instead of the full planned framework.

2. Background and Motivation

2.1 Stenosis and the quantities being modeled

Stenosis narrows the available flow area. Blood accelerates through the throat, pressure is lost across the lesion, and a separated or slowly moving region may form downstream. The exact response depends on severity, lesion length and shape, vessel diameter, flow rate, rheology, and the surrounding vascular system. Zhou et al. provide a useful idealized example in which increasing area blockage produces larger pressure loss, wall shear stress, and post-stenotic separation [2].

The primary fields in this project are velocity and pressure because they are the main fluid variables. Other quantities will usually be derived from them:

- Pressure drop is calculated between selected upstream and downstream locations.
- Wall shear stress is calculated from velocity gradients at the wall.
- Recirculation is identified from reversed or separated velocity regions.
- Oscillatory shear index can be calculated when a time-resolved pulsatile solution is available.

Keeping velocity and pressure as the main learned fields reduces the number of independent targets and makes derived quantities traceable to the same predicted flow solution.

2.2 Why begin with a first-principles model

A simple model is not automatically a poor model. Analytical, zero-dimensional, and one-dimensional cardiovascular models can represent conservation, vessel resistance, pressure-flow relationships, and pulse propagation at very low cost [17]. They are useful because the main structure of the problem is already known; a neural network does not need to rediscover all of it from examples.

Their weakness is local detail. A one-dimensional model may predict a pressure trend but cannot directly describe a three-dimensional jet or recirculation zone. An analytical axisymmetric profile may be accurate away from the lesion but inaccurate near an irregular throat. The proposed correction model is meant to focus on this gap.

2.3 POD as a compression method

POD finds a small set of spatial patterns that explain most of the variation in a collection of flow fields. A velocity field can then be approximated as a weighted sum of those patterns:

$$u(x,t;\mu) \approx \bar{u}(x) + \sum a_i(t;\mu) \phi_i(x) \quad (2)$$

The spatial modes ϕ_i are stored once. A new solution is described by the much smaller coefficient vector a . POD is useful only when the field family is genuinely low dimensional. Singular-value decay and held-out reconstruction error will be measured before any neural network is treated as justified.

2.4 Galerkin projection and physics-informed learning

Galerkin projection inserts the POD approximation into the governing equations and projects the result onto the same basis. This replaces the full discretized equations with a smaller system for the POD coefficients. Hijazi et al. combined this type of reduced system with a PINN and used it to estimate unknown parameters in inverse Navier-Stokes problems [1].

In this project, the neural network will not be trusted simply because a physics term is present in the loss. Its reconstructed fields, reduced residuals, conservation behavior, and inverse estimates must all be checked against held-out reference data. A low equation residual alone is not enough.

2.5 Why learn the discrepancy instead of the full field

The revised model assumes that the baseline already captures much of the ordinary flow. The learning problem is then changed from “predict the whole solution” to “predict what is missing.” If the remaining structure is smaller and more repeatable, the correction may need fewer POD modes and fewer high-fidelity cases. If not, the split adds complexity without benefit.

$$\delta q = q_{\text{HF}} - L(q_0), \quad \hat{q} = L(q_0) + \Phi \delta \hat{\delta} \quad (3)$$

Here q contains velocity and pressure, q_0 is the baseline, L maps the baseline into the field representation used by the high-fidelity data, δq is the discrepancy, $\Phi \delta$ is the discrepancy POD basis, and $\hat{\delta}$ is the predicted reduced coefficient vector.

2.6 Geometry is part of the model, not only an input number

Changing stenosis severity or location changes the shape of the vessel and therefore the mesh on which a flow solution is stored. POD cannot directly combine unrelated vectors unless the spatial points correspond. Geometry-informed registration, a common deformable mesh, reference-domain mapping, or local bases may be needed. Ye

et al. demonstrated one geometry-informed snapshot approach for stenosis and bifurcation examples [6]. This project will treat geometric correspondence as a measured source of error, not as an invisible preprocessing step.

2.7 Broader engineering pattern

Although arterial stenosis is the application studied here, the underlying pattern is broader: start with a fast model based on known equations, then learn only the systematic part that the model cannot resolve. Similar ideas can be applied to coarse CFD corrections, turbulence or combustion closures, thermal networks, structural reduced models, and other systems where a useful low-order model already exists. The stenosis case is suitable because both local flow detail and fast inverse estimation matter, but the project will remain focused on this one application during implementation.

3. Proposed Model Framework

3.1 Model hierarchy

The proposed workflow has five layers. Each layer should be testable independently before the complete system is assembled.

Layer	Purpose
1. Patient or virtual-case inputs	Geometry, flow conditions, rheology, available measurements, and a mask indicating missing values.
2. First-principles baseline	A fast model predicts the main pressure-flow behavior and, where possible, an approximate field.
3. High-fidelity discrepancy data	CFD, experimental, or other trusted data define the difference between the baseline and the reference.
4. Reduced correction and inverse model	POD compresses the discrepancy; a PINN estimates its coefficients and selected unknown parameters.
5. Reconstruction and checking	The correction is added to the baseline; velocity, pressure, derived metrics, uncertainty, and runtime are evaluated.

Figure 1. Conceptual model hierarchy. The word “figure” refers to this structured workflow table rather than a data-driven plot.

3.2 Selecting the baseline

The baseline must be simple enough to evaluate quickly but accurate enough that the correction remains manageable. The first study should compare at least two candidates rather than selecting one by convenience.

Candidate	Strength	Main limitation
Analytical or quasi-analytical stenosis model	Very fast; clear parameter dependence; useful for steady axisymmetric cases.	Limited geometry and local flow detail.
One-dimensional arterial model	Represents pressure and flow along a network and can support pulsatile cases.	Requires a field reconstruction before a velocity/pressure discrepancy can be formed.
Coarse axisymmetric or wedge CFD	Produces compatible fields and includes more local physics.	More expensive and less clearly separated from the high-fidelity model.
Lumped zero-dimensional model	Useful for boundary conditions, resistance, and system-level pressure-flow behavior.	Does not directly provide local spatial fields.

The choice may eventually be hierarchical. For example, a zero- or one-dimensional model could provide system-level boundary conditions while a low-resolution axisymmetric solution provides the field baseline near the lesion.

3.3 Building discrepancy snapshots

1. Evaluate the baseline and the high-fidelity reference at the same physical case.
2. Map both results into a common coordinate or mesh representation.
3. Subtract baseline velocity and pressure from the reference fields.
4. Store the original fields and the discrepancy fields so that full-field and corrected models can be compared.
5. Compute POD modes separately for velocity discrepancy and pressure discrepancy.
6. Measure how many modes are needed for a chosen reconstruction error.

A change to the baseline does not destroy the high-fidelity dataset, but it does require the discrepancy fields and their POD basis to be recomputed.

3.4 Physics of the combined field

The correction is not an independent fluid flow. The Navier-Stokes equations are nonlinear, so the equation evaluated on baseline plus correction is not equal to the baseline equation plus a separate correction equation. In compact form:

$$N(u_0 + \delta u, p_0 + \delta p) \neq N(u_0, p_0) + N(\delta u, \delta p) \quad (4)$$

The initial implementation should therefore evaluate the physical residual on the combined corrected field, or derive a consistent perturbation equation that keeps the interaction terms. This point matters because an incomplete split could make the model appear physically informed while enforcing the wrong equation.

3.5 Inverse formulation

The forward model maps known geometry and physiological parameters to a predicted flow. The inverse model goes in the other direction: it receives measurements and estimates the hidden parameters that could have produced them.

$$\text{observations } y_{\text{obs}} \rightarrow \text{estimated parameters } \hat{\mu} \text{ and corrected fields } \hat{q} \quad (5)$$

The first inverse problem should estimate stenosis severity while all other inputs are known. Later tests may add unknown lesion length, location, inlet flow, or rheology. The model should not be asked to infer several weakly identifiable parameters at once before the simple case works.

3.6 Observations, missing values, and uncertainty

Possible observations include pressure at selected locations, flow rate, Doppler velocity, partial velocity fields, or image-derived geometry. Systemic blood pressure can provide context but is not the same as local pressure loss across a lesion. The exact observation set will be chosen through identifiability experiments rather than assumed in advance.

Missing values should be represented explicitly with an observation mask. The first implementation may use a simple mask-aware neural input; more advanced imputation or probabilistic models can be added only if needed. When different lesions fit the same measurements, the output should show an interval or distribution instead of a single overconfident estimate.

3.7 Primary and derived outputs

Category	Outputs	Treatment
Primary fields	Velocity and pressure	Reconstructed from the baseline plus predicted discrepancy.

Category	Outputs	Treatment
Primary inverse parameters	Stenosis severity; later length and location	Estimated from available observations with uncertainty.
Derived hemodynamics	Pressure drop, WSS, recirculation, OSI when time-resolved	Calculated from reconstructed fields using documented post-processing.
Reliability	Prediction interval, out-of-domain indicator, correction magnitude	Used to identify ambiguous or unsupported cases.
Computational value	Online time and speedup	Reported with preprocessing and reconstruction costs separated.

4. Virtual Cohort and Data Strategy

4.1 Why a virtual cohort is being considered

A useful inverse model must see more than a few hand-picked stenosis cases. A virtual cohort would provide many plausible combinations of anatomy, physiology, disease, and measurement availability. It would support training, sensitivity studies, uncertainty analysis, and selection of the relatively small number of cases that need expensive CFD.

The goal is not to approximate every possible patient. The cohort will represent a stated population range and will inherit the limits of its data sources and assumptions. Jones et al. showed that a large virtual arterial population can be built from reported measurements, statistical relationships, and physiological constraints, and later used it to study stenosis and aneurysm detection from hemodynamic measurements [7,8].

4.2 Types of information needed

Variable group	Examples	Likely source
Geometry	Vessel diameter, lesion severity, length, location, eccentricity, curvature	Imaging datasets, published studies, parameterized geometry models.
Physiology	Flow rate, heart rate, systemic pressure, downstream resistance, viscosity	Population reports, authorized records, 0D/1D model calibration.
Measurements	Doppler velocity, pressure points, partial 4D-flow MRI, image-derived dimensions	Public research datasets or later authorized clinical collaboration.
Hemodynamic reference	Velocity and pressure fields, WSS, pressure loss	Public CFD, locally generated CFD, FDA benchmark data, experiments.
Metadata	Units, modality, measurement location, uncertainty, missingness	Dataset documentation and acquisition reports.

4.3 Public-data reality

Public resources exist, but they usually cover only part of the required pipeline. Geometry-informed blood-flow ROM data can support the spatial mapping problem [6]. A public aortic-stenosis dataset provides patient-derived geometric sequences and pressure/WSS targets for a data-driven model [11,12]. ARCADE provides labeled coronary angiography images for vessel and stenosis tasks [13]. The FDA nozzle and pump benchmarks provide experimental velocity and pressure data for CFD validation [14]. None of these, by itself, supplies every variable required for the proposed inverse discrepancy model.

Resource	Useful for	Limitation for this project
Geometry-informed stenosis/bifurcation ROM study [6]	Geometry registration, compatible snapshots, direct ROM comparison.	Non-intrusive forward ROM; exact data availability and license require audit.
Aortic-stenosis training data [11,12]	Patient-derived geometric sequences and pressure/WSS prediction context.	Not a complete volumetric velocity-pressure dataset for this POD-Galerkin formulation.
ARCADE [13]	Image-based vessel and lesion localization.	No paired three-dimensional hemodynamic fields.
FDA flow benchmarks [14]	Independent CFD verification using experimental pressure and velocity.	Generalized device geometries rather than patient arterial stenosis.
Locally generated idealized cases	Complete control of mesh, velocity, pressure, rheology, and boundary conditions.	Limited realism and a local compute cost.

4.4 How the virtual cohort would be generated

The initial cohort should be built in stages. It should not begin by independently drawing every variable from a minimum and maximum value. Independent sampling can create combinations that are individually plausible but jointly impossible.

1. Define a data dictionary with units, source, expected range, and whether each variable is observed, sampled, simulated, inferred, or derived.
2. Use published distributions and authorized de-identified data, if available, to estimate relationships between geometry, physiology, and disease state.
3. Fit an interpretable joint model such as a hierarchical Bayesian model, Bayesian network, or copula before considering a more complex neural generator.
4. Sample candidate virtual cases and reject or repair cases that violate explicit geometry and physiology constraints.
5. Run the first-principles model for every accepted case.
6. Select a smaller, informative subset for high-fidelity simulation or use available reference fields.
7. Form discrepancy snapshots and train the reduced correction.
8. Generate simulated measurement noise and missingness only after the underlying case and reference output are defined.

4.5 Physical and statistical constraints

Examples of required checks include positive dimensions and viscosity, a throat smaller than the healthy lumen, a lesion located inside the modeled segment, consistent pressure definitions, plausible systolic and diastolic relationships, mass conservation, and compatible flow and Reynolds-number calculations. Rejection rates and reasons must be stored. A very high rejection rate would suggest that the joint population model is poorly specified.

Stenosis severity and hemodynamic outputs should not be generated as unrelated labels. A preferred direction is:

geometry + physiology → forward model → hemodynamic fields → simulated observations (6)

The inverse network then attempts to recover the hidden geometry and physiology from those observations. This keeps the synthetic data internally consistent even when the first-principles model is approximate.

4.6 Multi-fidelity sampling

A large virtual cohort does not require a high-fidelity three-dimensional simulation for every case. A practical hierarchy could contain tens of thousands of 0D/1D evaluations, a larger set of analytical or axisymmetric cases, a

smaller set of coarse CFD cases, and a carefully selected set of high-fidelity cases. Active learning may later select new CFD cases where the correction model is most uncertain or where the baseline error changes rapidly.

4.7 Training splits, missingness, and noise

If real patient-level data are used, the train, validation, and test split must occur before synthetic expansion. Derived records from the same person cannot appear in both training and testing. Missing measurements should initially be removed in controlled patterns and later matched to the observed dataset. Noise should be based on the measurement type; a pressure sensor, Doppler measurement, segmentation, and MRI velocity field do not have the same error model.

4.8 Limits of synthetic data

Synthetic expansion can improve coverage of a defined parameter space, but it cannot create relationships that are absent from the source information, remove bias from an unrepresentative cohort, or prove clinical performance. A generator trained on de-identified records may also reproduce rare records or be restricted by a data-use agreement. Public release of any patient-informed synthetic data would require a separate privacy and licensing review.

5. Study Design, Baselines, and Validation

5.1 Initial controlled parameter domain

$$\mu = [\eta, L_s/D, Re, \text{rheology}] \quad (7)$$

The initial model family will emphasize stenosis severity η , nondimensional lesion length L_s/D , Reynolds number Re , and a selected rheology description. Vessel diameter may also be included when it is not removed through nondimensionalization. Location, eccentricity, curvature, pulsatility, and multiple lesions are later extensions. They should not all be introduced before the basic discrepancy model is verified.

5.2 Required comparison models

1. First-principles baseline without any learned correction.
2. Full-field POD reconstruction, which measures the best approximation available from a selected basis.
3. Full-field POD interpolation or radial-basis regression.
4. Full-field data-only POD neural network.
5. Discrepancy POD reconstruction.
6. Baseline plus a data-only discrepancy neural network.
7. Baseline plus the POD-Galerkin PINN discrepancy model.
8. High-fidelity computational or experimental reference.

A model should not be retained because it is more sophisticated. The proposed method needs to provide a clear benefit in accuracy, data use, stability, speed, uncertainty handling, or physical consistency.

5.3 Verification before learning

- Geometry parameters reproduce the intended severity and lesion dimensions.
- The baseline is checked against analytical limits, published trends, or trusted numerical cases.
- The high-fidelity solver passes mesh, conservation, and convergence checks appropriate to the selected case.
- Baseline and high-fidelity fields are mapped into the same space without a hidden ordering or unit error.
- POD reconstruction is verified independently of the neural network.
- Derived pressure drop and WSS are computed from documented locations and wall definitions.

5.4 Core validation metrics

Metric	What it answers
Relative velocity and pressure error	Can the corrected model reconstruct the primary fields?
Severity and location error	Can the inverse model recover the lesion parameters?
Pressure-drop and WSS error	Are important derived outputs preserved?
Baseline error removed	How much value does the correction add over the simple model?
Correction magnitude	Is the correction small enough that the baseline still carries meaningful structure?
POD mode count and singular-value decay	Is the discrepancy actually easier to compress than the full field?
Training-data requirement	Does the physics-informed model reach a target accuracy with fewer high-fidelity cases?
Inference time and speedup	Does the online model provide a practical computational advantage?
Noise and missing-data sensitivity	How quickly does performance degrade as measurements become less complete?
Uncertainty calibration	Do reported intervals reflect the frequency of actual errors?

5.5 Stage-gated experiments

Gate	Question	Evidence required
A. Baseline	Does a cheap model capture the main behavior?	Comparison with several known stenosis cases and clear error maps.
B. Discrepancy POD	Is the discrepancy compact?	Mode decay and held-out reconstruction versus full-field POD.
C. Forward correction	Can the reduced model correct unseen cases?	Held-out velocity, pressure, and derived-output error.
D. Inverse severity	Can severity be inferred when other inputs are known?	Multiple initializations, noise tests, and held-out cases.
E. Missing data and uncertainty	Does the model recognize weaker evidence?	Sensor ablation and calibrated prediction intervals.
F. Added geometry	Does the approach survive more variation?	Held-out length, location, or eccentricity cases.
G. Demonstration release	Is the workflow reproducible?	Documented examples, fixed environments, tests, and license review.

5.6 Narrowing and fallback conditions

The project should be narrowed when the available evidence does not support the larger architecture. If the discrepancy is not compact, the project may return to a full-field POD model. If severity cannot be inferred from the selected observations, the focus may shift to measurement design or forward prediction. If public data are insufficient, the first paper-quality result may be a controlled idealized benchmark rather than a patient-informed model.

6. Feasibility and Computing Resources

Reduced-coordinate neural training is expected to be practical on the available workstation because the network predicts tens or hundreds of coefficients rather than millions of mesh values. The expensive part is high-fidelity

data generation and storage. The initial project should therefore favor public data, analytical and 1D models, two-dimensional axisymmetric cases, or a thin OpenFOAM wedge before attempting a large patient-specific three-dimensional campaign.

A symmetry reduction is valid only when the geometry, inlet, outlet, wall behavior, and expected flow remain symmetric. Eccentric lesions, curved vessels, bifurcations, and asymmetric separation may require a full three-dimensional model. Institutional computing may become useful later, but the first feasibility gates should be designed to run with personal resources.

The main non-computational constraint is likely to be data alignment: finding paired geometry, physiological measurements, and trustworthy reference fields with clear permissions. This is why the data audit and small pilot study come before a large software build.

7. Main Risks and Mitigation

Risk	Why it matters	Response
Baseline is too crude	The correction must reproduce nearly the whole flow, defeating the split.	Compare baseline candidates; restrict the domain; use a slightly higher-fidelity baseline.
Discrepancy is not low dimensional	Many POD modes reduce speed and stability.	Measure before training; use local bases or return to full-field POD.
Baseline-to-field lifting creates artifacts	A 0D/1D result may not map cleanly into a 2D/3D field.	Test the lifting separately and include its error in validation.
Nonlinear interaction is omitted	The physics loss may enforce an incorrect correction equation.	Evaluate residuals on the combined field or derive the perturbation terms carefully.
Geometry mapping dominates error	POD may learn registration artifacts instead of flow behavior.	Use common topology where possible and report mapping error separately.
Inverse parameters are not identifiable	Different lesions may match the same observations.	Add measurements, infer a distribution, or redefine the task around identifiability.
Virtual cohort contains false correlations	The model may learn synthetic artifacts.	Use joint distributions, physical constraints, held-out real/reference tests, and sensitivity studies.
Synthetic-to-real gap	Good synthetic results may not transfer to patient data.	Keep claims within the simulated domain and require independent later validation.
Correction overwhelms physics	The model becomes a data-driven predictor with a decorative baseline.	Track correction magnitude, regularize it, and compare with a direct neural baseline.
Scope becomes too large	The cohort, CFD, inverse model, uncertainty, and interface may be too much at once.	Use the stage gates and stop after a defensible smaller study if necessary.

8. Scope and Claim Boundaries

8.1 Included in the initial research scope

- A controlled family of idealized or registered arterial stenosis geometries.
- A first-principles baseline and a reduced correction built from velocity and pressure discrepancies.
- Forward reconstruction and inverse estimation of stenosis severity.
- Comparison with full-field and data-only reduced models.
- Synthetic noise, missing observations, and uncertainty experiments.
- A physiologically constrained virtual cohort, initially based on public aggregate information and simulation.

- Open-source code and a demonstration interface after verification.

8.2 Excluded from the initial claims

- Clinical diagnosis, treatment recommendation, or patient-safety claims.
- Universal performance across arbitrary arterial anatomy.
- Guaranteed superiority over simpler ROM or machine-learning models.
- Greater fidelity than the high-fidelity reference used to train or test the correction.
- Automatic inference of local stenosis from systemic blood pressure alone.
- Fluid-structure interaction, vessel growth, or a complete cardiovascular network in the first implementation.
- Public release of patient-derived data without explicit permission and privacy review.
- A finished clinical or commercial product.

9. Research Practice, Documentation, and Open-Source Intent

9.1 Independent research record

The project will be documented through the repository, this project-definition document, and dated engineering-notebook entries. Public literature, public or authorized datasets, original code, and personal computing resources will be used. Each important model choice should be connected to a notebook entry and a reproducible configuration rather than preserved only in conversation or memory.

9.2 AI-assisted development

Generative AI tools are being used to accelerate drafting, code scaffolding, source discovery, debugging, and routine organization. They are not treated as a source of authority. Equations, citations, design decisions, software behavior, and results remain the responsibility of the researcher and must be checked through original sources, code review, tests, and benchmark comparisons. Tracking every AI interaction is not required, but important AI-assisted decisions or generated code that affects results should be reviewable through commits and notebook records.

9.3 Open-source plan

The current repository may remain private while the method is being defined and permissions are reviewed. A public release should occur only after the code has a reproducible example, licenses have been checked, nonredistributable data have been removed, and the documented claims match the available evidence. The interface is a demonstration layer, not the scientific contribution by itself.

9.4 Document and repository versioning

The recommended repository policy is to maintain one canonical current file, for example ``docs/research_project_definition.docx`` or its PDF export, and keep milestone versions in ``docs/archive/``. A milestone version should be retained when the hypothesis, model architecture, scope, or project status changes. Git commits preserve routine wording and formatting edits, so a separate file is not needed for each small revision.

10. Immediate Next Actions

1. Archive version 0.1 and adopt version 0.2 as the current project definition.
2. Create a literature and dataset matrix covering available fields, geometry, patient variables, licenses, and access requirements.
3. Write a small comparison table for candidate first-principles baseline models.
4. Choose one simple idealized stenosis benchmark and reproduce its main pressure and velocity trends.
5. Define the first virtual-case schema with units and classify each variable as sampled, observed, simulated, inferred, or derived.

6. Build a small pilot dataset before attempting a large virtual cohort.
7. Compute full-field and discrepancy POD bases on the pilot cases and compare their mode decay and reconstruction error.
8. Continue to a neural correction only if the discrepancy representation offers a measurable reason to do so.
9. Define the simplest inverse severity experiment and the exact observations it will receive.
10. Record all changes, failed assumptions, and baseline-selection results in the engineering notebook.

11. Conclusion

The project now centers on a simple idea: use known hemodynamic equations for the part of the problem they already describe well, and use a reduced learned correction for the remaining error. POD provides a way to test whether that error is compact. Galerkin projection and the PINN provide a way to connect the reduced correction to the governing physics and to estimate unknown stenosis parameters. A virtual cohort provides a controlled population on which the inverse problem, missing measurements, and uncertainty can be studied.

The value of the work will depend on evidence, not on the sophistication of the architecture. The first decisive result is whether the discrepancy is actually easier to represent than the full field. The next is whether the correction improves unseen cases without becoming larger than the baseline or hiding a non-identifiable inverse problem. If those tests are passed, the approach can grow toward more realistic geometry and patient-informed data. If they are not, the documented comparison will still establish a useful boundary for the method.

Declaration of Generative AI and AI-Assisted Technologies

Generative AI tools were used during preparation of this document for drafting support, organization, source discovery, and document-production assistance. The author reviewed and revised the content, checked the cited sources used to support technical claims, and remains responsible for the project definition. Future code, analyses, and manuscripts will require independent testing, verification, and disclosure consistent with the applicable venue.

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Appendix A. Symbols and Initial Parameter Definitions

Symbol	Meaning	Current role
η	Area-based stenosis severity	Primary initial inverse parameter.
L_s/D	Stenosis length divided by reference diameter	Initial geometric parameter.
Re	Reynolds number	Represents the balance of inertia and viscosity.
q_0	Baseline velocity and pressure representation	Fast first-principles prediction.
q_{HF}	High-fidelity reference field	CFD or trusted experimental/computational reference.
$L(\cdot)$	Lifting or mapping operator	Places the baseline in the same field space as the reference.
δq	Reference minus lifted baseline	Field learned by the reduced correction.
$\Phi \delta$	Discrepancy POD basis	Spatial modes for the correction.

Symbol	Meaning	Current role
$a\delta$	Discrepancy coefficients	Predicted reduced state.
y_{obs}	Available observation vector	Pressure, flow, velocity, or geometry measurements.
m	Observation mask	Indicates which measurements are present.
WSS / OSI	Wall shear stress / oscillatory shear index	Derived outputs; OSI requires time-resolved data.

Appendix B. Initial Metric Definitions

$$\text{relative field error} = ||q_{\text{reference}} - q_{\text{predicted}}||_2 / ||q_{\text{reference}}||_2 \quad (\text{B1})$$

$$\text{severity error} = |\eta_{\text{true}} - \eta_{\text{predicted}}| \quad (\text{B2})$$

$$\text{fraction of baseline error removed} = 1 - (\text{corrected error} / \text{baseline error}) \quad (\text{B3})$$

$$\text{speedup} = \text{high-fidelity wall time} / \text{total online evaluation time} \quad (\text{B4})$$

Pressure errors should account for arbitrary reference offsets where applicable. WSS should be reported with more than one measure because a small global field error can still hide a large local gradient error. Uncertainty evaluation should report how often the true value falls inside the stated interval and how wide that interval is.

Appendix C. Decision Record

Decision	Current position	Reason for later review
001 - Private start	Begin in a private repository.	Public release follows verification and license review.
002 - Documentation first	Define the project before core solver development.	The definition can be revised after feasibility tests.
003 - Baseline plus discrepancy	Use a first-principles model as the main prediction and learn the remaining error.	Continue only if discrepancy POD shows a measurable advantage.
004 - Virtual cohort	Investigate a physiologically constrained synthetic population.	Method depends on available data and validation.
005 - Preserve uncertainty	Do not assume the revised architecture is superior.	Negative results may return the project to a simpler full-field model.
006 - Version milestones	Archive major research-definition versions; update the canonical file for minor edits.	Git history already records routine changes.

Appendix D. Documentation Hierarchy

Record	Purpose	Version practice
Current RPDD	Latest stable statement of the project idea, scope, method, and claim boundaries.	Update in place between major changes.
Archived RPDD milestones	Preserve major changes in hypothesis or architecture.	Keep v0.1, v0.2, and later milestone releases only.
Engineering notebook	Dated ideas, experiments, failed attempts, decisions, and immediate next steps.	New entry for each substantial work session or decision.

Record	Purpose	Version practice
Software documentation	Installation, data manifests, configurations, interfaces, and reproducible examples.	Version with the code.
Technical report or paper	Final method, evidence, limitations, and interpretation after research is completed.	Created only when results support it.